

SAAKSHI - ???????, the witness

AI-powered CCTV integration and video analytics for Gujarat Police

- 26 departments, ~80,000 cameras, and no single place that knows what exists or whether it works.
- We submit under **Model 1 (compulsory) + Hybrid**, mapped onto both of the portal's model numberings.
- Every figure in this deck is measured on the government sandbox feeds and traceable to a ticket. Where the system misses a stated target, the slide says so.

The limits are stated first, on purpose. In a room of overpromising vendors, the team that states its limits is the one believed on everything else.

Model choice - and the numbering contradiction we resolved ourselves

- The portal carries **two model definitions that disagree**. /problems calls Model 3 "VMS Federation & Middleware" - software. The unlinked /evaluation-criteria page scores Model 3 as **hardware**: transponder/encoder, secure boot, PoE, rugged design.
- /problems Step 3 explicitly permits "a hybrid architecture combining features from two or more reference solution models". The helpdesk never resolved the numbering, so we used that permission rather than wait.
- **We submit as Model 1 (compulsory) + Hybrid**, and map our deliverables onto both numberings: under /problems our adapter framework is Model 3; under /evaluation-criteria our analytics and alerting are Model 4.
- A scorer applies whichever rubric they hold and still finds the work mapped to it. **We build no hardware, and we say so plainly.**

Model 1 is compulsory and is satisfied under either reading.

Rubric mapping - Model 1 (Registry & GIS)

SCORED LINE	MARKS	OUR ANSWER
Data Accuracy & Registry Completeness	25	Registry + trust score - measured, never declared
GIS Visualization & Usability	20	MapLibre registry map, trust overlay, gap analysis
API Design & Integration Readiness	20	OpenAPI registry API + docs/registry-api.md
Scalability & Performance	20	Edge-metadata architecture; measured sizing model
Security & Access Control	10	RBAC + hash-chained, verifiable audit chain
Innovation & Value Addition	5	Trust score, retention clock, gap analysis

Rubric mapping - Model 4 (Centralized Analytics & AI Insights)

SCORED LINE	MARKS	OUR ANSWER
System Architecture & Integration Depth	25	Adapter framework unifying cameras, registry, control room
Core Functional Modules & Workflow Automation	25	Discovery, connectivity, status, health, uptime, alerting
Reliability & Scalability	20	Edge analytics; backhaul arithmetic; sizing model
Security & Access Control	20	RBAC, purpose binding, audit chain, export bundles
Innovation	5	Impossible-transition detection; plate-cloning evidence
API Ecosystem	5	OpenAPI, typed client, documented onboarding

On /evaluation-criteria the models read as complementary layers, not four alternatives - Model 4 is explicitly scored on how well it "integrates Cameras, Registry (M1) and Control Room (M2)".

Overall architecture

- **Video stays where it is.** Cameras remain owned and operated by their department. We add a registry, an adapter layer and an analytics plane - not a new central VMS.
- **Adapter framework** normalises HLS, RTSP, ONVIF, WHEP and NVR sources behind one interface, so a heterogeneous estate presents a single contract.
- **Analytics at the edge**, metadata to the centre: detections, plate reads, crops and health signals travel; video does not.
- **PostgreSQL + PostGIS + TimescaleDB** for registry, sightings and time-series; **Valkey Streams** between workers; **MinIO** for evidence objects.
- **Every claim the product makes about a camera is measured by probing it**, because the government catalogue publishes {id, name} and nothing else.

Integration strategy - heterogeneous cameras, NVRs and VMS

- One `CameraAdapter` interface; five implementations (HLS, RTSP, ONVIF, WHEP, NVR). Adding a vendor is a new adapter, never a change to the core.
- **Capabilities are probed, not declared:** codec, resolution, frame rate and decodability are measured from the stream itself.
- RTSP forced over TCP, with HLS fallback when 8554 is blocked. Reconnect backoff 2s->30s - never a tight loop.
- **Timing comes from the frame's presentation timestamp (PTS), never arrival time.** A gateway that replays a buffered GOP on reconnect would otherwise produce impossible velocities.
- Bulk onboarding by CSV (`POST /api/v1/cameras/bulk`) for estates that arrive as a spreadsheet - which is how coordinates reach the system, because the catalogue carries none.

AI and video analytics

- YOLO11 detection + ByteTrack tracking + ONNX plate OCR, with **best-shot selection** and a multi-frame vote rather than a per-frame read.
- **ANPR is the only mandatory analytic.** Vehicle attributes, re-identification, route reconstruction and cloning detection are additional and each is reported with its own measurement.
- A read is a **string**, never an identification. The UI leads with that verdict before the severity.

MEASURED ON THIS ESTATE	VALUE
Exact read recall (reads equal to the human label, over legible plates)	0 of 3 - 0%
Precision (correct reads over all reads emitted)	0%
Plate-detection recall (plate boxes over human-legible plates)	100% on n=3
Character accuracy ($1 - \text{editDistance}/\text{len}$)	51.8%
Human-legible plates in the hand-labelled sample	3 of 120

Cybersecurity architecture

- **RBAC with four roles** (admin, supervisor, operator, auditor), enforced server-side. The browser's role cookie is a hint; every capability is re-checked against a signed token.
- **Purpose binding**: a vehicle trace or plate search without a stated purpose is rejected 400, server-side. The purpose and the officer's badge are written into the audit chain.
- **Hash-chained, append-only audit log**. The database grants SELECT and INSERT only; UPDATE and DELETE raise `restrict\_violation` even for the owner.
- **Evidence is session-gated**: crops are streamed by the API behind the session, and the object store has no public network surface.
- Bearer token in an httpOnly cookie - never reachable from browser JavaScript. Every published container port binds loopback by default.
- **No biometrics are processed.**

Deployment architecture

- **Two topologies, and the choice is a measurement, not a preference.** Where feeds are reachable from the datacentre, everything runs centrally. Where they are not - the sandbox case - the control plane is hosted and the ingest workers run beside the cameras.
- Fully containerised: ``docker compose up`` brings the whole stack up from a clean clone. One command, no manual steps.
- Edge nodes carry analytics; the centre carries registry, alerting, trace, audit and the console.
- All open source. The only proprietary dependency is an optional NL-query LLM behind a four-provider interface - with ``ollama`` or ``none`` the system is fully functional and fully open.

Infrastructure sizing - computed, not estimated

	PILOT - 500 CAMERAS	STATEWIDE - 80,000 CAMERAS
ANPR cameras	500	24000
Accelerators required	63	240
District nodes	33	33
Events per day	1,05,84,000	1,69,34,40,000
Retained storage	2-7 TB	841-3493 TB
Total backhaul	0.00 Gbps	0.06 Gbps

Every figure on this slide is computed by the same model behind the in-product sizing calculator, from constants each tagged measured, listed or assumed. Nothing here is typed by hand.

Cost-benefit analysis

	PILOT - 500 CAMERAS	STATEWIDE - 80,000 CAMERAS
Capex	INR 3.07 crore - INR 5.12 crore	INR 11.70 crore - INR 19.50 crore
Annual opex	INR 49.52 lakh - INR 83.46 lakh	INR 2.23 crore - INR 7.05 crore
Total annual cost	INR 1.11 crore - INR 1.86 crore	INR 4.57 crore - INR 10.95 crore
Annual cost per camera	INR 22,189 - INR 37,168	INR 571 - INR 1,369

The benefit is not a headcount saving. It is that a question an officer cannot answer today - "where did this vehicle go, and can I prove it" - becomes answerable in seconds, with an audit trail that survives cross-examination. Bands are quoted rather than point estimates because hardware pricing is listed, not measured.

Department-wise information requirements

- **What we need from each department to assess integration feasibility**, as a questionnaire rather than an assumption:
- Camera inventory: count, make/model, IP or analog, codec, resolution, frame rate, mount type, and whether coordinates exist.
- VMS/NVR in use: vendor, version, whether an API or ONVIF profile is exposed, and who holds the credentials.
- Network: uplink bandwidth per site, whether the site can reach a central endpoint, and any firewall constraint on RTSP.
- Retention: how long footage is kept today, on what storage, and who may request it.
- Governance: who authorises access, what purposes are permitted, and what must be logged.
- **Nobody else will ask these questions, and they are scored.** A feasibility answer without them is a guess.

Scalability strategy

- **The architecture scales because video does not move.** Adding cameras adds edge compute and metadata, not backhaul - which is the difference between a system that reaches 80,000 cameras and one that does not.
- Registry measured at **1,00,000 cameras and 500 concurrent connections with zero failed responses**, on a single Node process sharing a laptop with its own database.
- Keyset pagination: the deep cursor page (row ~90,000) is **faster** than the first page - 265 ms against 345 ms - so cost does not grow with offset.
- Analytics scales horizontally by district node; each node is sized from measured per-stream throughput, not a vendor datasheet.
- TimescaleDB hypertables for sightings and health checks; retention enforced by policy rather than by a cron nobody owns.

Future roadmap

- **Near term - the gaps we found and stated.** Live VAHAN/SARTHI/eGujCop connectors behind the existing provider interface, replacing the mock. Plate-model retraining on Indian plates at the geometry this estate actually presents.
- **Accuracy.** The estate is dominated by wide-area traffic-overview cameras; the roadmap is siting guidance from the gap analysis plus per-camera model selection, not a promise that the same model will do better.
- **Coverage.** The gap analysis already names the junctions with zero trusted coverage; the roadmap turns that into a procurement plan.
- **Federation breadth.** More NVR/VMS adapters as departments are surveyed, each with a conformance test against a real device.
- **Vehicle re-identification** ships disabled at 0.761 held-out precision against a 0.9 bar. It is enabled when a measurement clears the bar, not before.

The backhaul argument - 125x, as arithmetic

AT 80,000 CAMERAS	BANDWIDTH
Every camera streamed centrally - Model 4 as /problems describes it	160 Gbps
Metadata from cameras analysed at the edge	1.28 Gbps
Total under our architecture	1.28 Gbps

Reduction: 125x

A fully centralised VMS is not merely expensive - it requires 26 departments to surrender infrastructure they own and operate. Our Model 4 is analytics over a federated estate, and the video never leaves the department that owns it.

This is the conservative figure: it runs on the ORIGINAL assumed event rate of 2,000 B/s so that it reproduces PROJECT.md section 9 exactly and can be checked against it. Our measured per-track event rate is far lower, which takes the same estate to 2848x. We quote the smaller number.

The registry that tells the truth

- **The government catalogue publishes** `{"id": "cam01", "name": "01 Chiman bhai Bridge"}` **and nothing else.** No codec, no frame rate, no resolution, no coordinates. So every property in our registry is **measured by probing the stream.**
- A **trust score** per camera, from connectability, decodability, focus and clock drift - and a camera that has never been probed resolves to `band: null`, which is an absence of evidence and is displayed as one.
- Measured on this estate: **6 resolutions, 6 frame rates, and a 24x spread in recording duration** across 30 cameras that declare none of it.
- **Two cameras in the same city and the same hour returned 67 and 33,548 sightings** - a 500x spread. The quiet one is not broken: it decoded 5,582 frames cleanly at 23.16 fps. It simply sees almost no vehicles. A map of where cameras *are* cannot show that; a registry that measures them can.

Coverage you can defend, and evidence with a clock

- **Every metre of apparent coverage in this estate is contributed by a camera nobody has verified.** All-camera coverage is 21.47 km of a 218,137 km road network; **trusted-only coverage is 0.00 km** - a delta of 100%.
- A conventional coverage map would have drawn all 21.47 km in green. Ours draws the distinction, and names the **6,750 junctions with zero trusted coverage**.
- **The retention clock.** Footage on this estate is kept 7-15 days. The system tracks, per piece of evidence, how long it has left - and lets an officer preserve it before it expires. Nobody tracks this today, and it is the difference between evidence and a story about evidence.

Impossible transitions, and a chain of custody

- **Plate cloning, detected as physics.** Two sightings of one registration 9.24 km apart with 30 seconds between them demand 1,109 km/h. The route is reconstructed on a real 540,711-way road graph, so "impossible" is a road distance and a travel time - not a threshold someone chose.
- The tool refuses to overclaim in both directions: with no road distance it reports **"0 transitions were testable", not "the estate is clean"**, and when the impossible rate is implausibly high it says to read it as an OCR or clock finding first.
- **Chain of custody.** Every search, export and alert transition is written to a hash-chained append-only audit log carrying the officer's badge and their stated purpose.
- **Export bundles are independently verifiable** - crops embedded as bytes, a manifest hashed, and a `verify.mjs` that runs with nothing but Node. A signed URL is never written into a bundle, because it expires and then looks real while being useless.

The stated benchmarks - all six answered

TARGET	MEASURED	VERDICT
1,00,000+ camera records	1,00,000 cameras benchmarked	meets
API response < 200 ms	p95 110 ms @200 conns; 252 ms @500	meets to 200; over at 500
Dashboard load < 3 s	readable 132 ms; map complete 1.65 s	meets
Detection accuracy > 90%	exact read recall 0%	MISSES
Uptime > 99%	100.000% over a stable 30 min	not measured over a meaningful period
500+ concurrent users	500 concurrent, zero failed responses	meets on throughput

Two misses and one unmeasured, printed here rather than omitted. A deck that silently drops the rows it fails is the one a scorer stops trusting - and every figure above is reproducible from the command recorded against it in docs/claims-provenance.md.

What this system does not do

- **No live VAHAN / SARTHI / eGujCop / AFIS / NAFIS connection.** A mock provider and a written connector specification. We do not claim access we were never given.
- **No face recognition.** Deliberately out of scope: not mandated, and it requires separate legal authorisation. We process no biometrics.
- **No central video storage.** The architecture refuses it, and the arithmetic is on slide 15.
- **No VLM "suspicious activity detection."** The cost and latency at 80,000 cameras do not survive procurement; "suspicious" is not a definable class, so no accuracy claim about it is falsifiable; and "the model said it looked suspicious" is not evidence in court. A plate read with a timestamped crop and a confidence score is.
- **No accuracy claims without measurement.** We report measured precision and recall including where the system fails - night, two-wheelers, oblique angles, motion blur.